School of Engineering Technology

Main Campus, Off Hennur-Bagalur Main Road, Chagalahatti, Bengaluru-562149

MINI PROJECT REPORT

"FAKE PROFILE DETECTION SYSTEM"

submitted to,

School of Engineering and Technology, CMR University

in partial fulfilment of the requirement for the award of the degree of

Bachelor of Technology,
in
COMPUTER SCIENCE AND ENGINEERING(AI&ML)

by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING-(AI&ML)
CMR UNIVERSITY

2025-26

School of Engineering and Technology, CMR

Main Campus, Off Hennur-Bagalur Main Road, Chagalahatti, Bengaluru-562149

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING-(AI&ML)

CERTIFICATE

*Certified that the mini project titled **Fake Profile Detection System** carried out by Mr./Ms. **Kavya Shree V (24BBTCA060), Neysa Mary Pramod (24BBTCA078), Pooja Rajpurohit (24BBTCA084)** in partial fulfilment for the award of Bachelor of Technology in **COMPUTER SCIENCE AND ENGINEERING-** (AI&ML) of CMR University, during the year 2025-26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The mini project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.*

Examiners

Signature of the Guide

Prof. Gripsy Paul
Assistant Professor
Dept. of CSE, SoET,
CMRU, Bangalore

Name of the examiners

1.

Signature with date

DECLARATION

We, **Kavya Shree V, Neysa Mary Pramod, Pooja Rajpurohit** students of School of Engineering and Technology, CMR university, hereby declare that the dissertation titled **"Fake Profile Detection System"** embodies the report of my mini project carried out independently by us during fourth semester of **Bachelor of Technology in Computer Science and Engineering(AI&Ml),** under the supervision of **Prof. Gripsy Paul,** Department of Computer Science and Engineering and this work has been submitted in partial fulfilment for the award of the **Bachelor of Technology** degree.

We have not submitted the project for the award of any other degree of any other university or institution.

Date :

Place :

Kavya Shree V (24BBTCA060)
Neysa Mary Pramod (24BBTCA078)
Pooja Rajpurohit (24BBTCA084)

1. Abstract

The rapid growth of social media platforms has led to an increase in fake user profiles, which are often used for malicious purposes such as spreading misinformation, scams, and spamming. Identifying such fake profiles manually is a challenging and time-consuming task. This project aims to develop an Artificial Intelligence (AI) based system that can automatically detect fake profiles using machine learning techniques.

The system uses profile-related features such as number of followers, number of following, number of posts, bio length, and account age to classify whether a profile is real or fake. A machine learning model based on the Random Forest Classifier is trained using a labeled dataset. The dataset is preprocessed, and the model is evaluated using performance metrics such as accuracy and confusion matrix.

The system is implemented using Python and deployed as a web-based application using Streamlit, which provides an interactive and user-friendly interface. The results indicate that the model can effectively classify fake and real profiles with satisfactory accuracy. This project demonstrates the practical application of machine learning in solving real-world cybersecurity problems.

Furthermore, the system is designed to be scalable and adaptable, allowing it to handle large volumes of user data efficiently. Future enhancements may include incorporating additional features such as user activity patterns, content analysis, and network behavior to further improve detection accuracy. The model can also be integrated with social media platforms for real-time monitoring, thereby strengthening security measures and reducing the impact of fraudulent accounts.

2. Introduction

In today's digital world, social media platforms such as Instagram, Facebook, and Twitter have become an integral part of communication and information sharing. However, the increasing popularity of these platforms has also led to the rise of fake profiles. These fake accounts are often created to perform malicious activities such as spreading fake news, phishing, online fraud, and increasing follower counts artificially.

Detecting fake profiles is a critical challenge faced by social media companies. Traditional methods of detection involve manual verification or simple rule-based systems, which are not efficient for handling large volumes of data. With the advancement of Artificial Intelligence and Machine Learning, automated systems can now be developed to detect such profiles more efficiently.

This project focuses on designing and implementing a machine learning-based system for detecting fake profiles. The system analyzes user behavior and profile characteristics to make predictions. The significance of this project lies in enhancing online security and improving the reliability of social media platforms.

The scope of this project includes developing a classification model, integrating it with a user interface, and demonstrating its functionality. The report is organized into sections covering problem statement, objectives, system analysis, design, implementation, results, and conclusion

3. Problem Statement

The increasing number of fake profiles on social media platforms poses a serious threat to users and organizations. These fake accounts are often used for fraudulent activities, identity theft, spreading misleading information, and executing cyber- attacks. Many of these profiles are designed to closely resemble genuine accounts, making them difficult to identify. As social media platforms continue to grow rapidly, the scale and impact of such malicious activities have also increased significantly.

Manual detection of fake profiles is inefficient, time-consuming, and prone to errors. It requires continuous monitoring and human effort, which is not feasible given the vast number of users and the dynamic nature of online platforms.
Additionally, traditional rule-based approaches fail to adapt to new and evolving patterns of fake account behavior, resulting in reduced detection accuracy and effectiveness.

Therefore, there is a need for an automated system that can accurately and efficiently detect fake profiles using data-driven techniques. Machine Learning (ML) provides a powerful approach to address this challenge by analyzing large volumes of data and identifying hidden patterns that distinguish fake profiles from real ones. By leveraging user profile attributes and behavioral features, ML models can make reliable predictions with minimal human intervention.

This project aims to solve this problem by developing a machine learning-based system that can classify profiles as real or fake based on their characteristics. The system considers various features such as number of followers, following count, posting activity, profile completeness, and engagement patterns to improve classification accuracy. Advanced algorithms are used to train and test the model, ensuring that it performs effectively on unseen data.

The proposed solution benefits social media users by enhancing their security and protecting them from fraudulent interactions. It also assists platform administrators in identifying and monitoring suspicious activities more efficiently. Furthermore, the system contributes to improving trust and reliability in online platforms by reducing the presence of fake accounts. Overall, this project demonstrates how machine learning can be applied to address real-world cybersecurity challenges in a scalable and effective manner.

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4. Objectives

The primary objective of this project is to develop an AI-based system for detecting fake profiles using machine learning techniques. The specific objectives of the project are as follows:

- To collect and preprocess profile-related data for training the model.
- To design and implement a machine learning model capable of classifying profiles as real or fake.
- To train the model using a suitable algorithm such as Random Forest.
- To evaluate the performance of the model using metrics such as accuracy and confusion matrix.
- To develop a user-friendly web interface using Streamlit for easy interaction.
- To display prediction results along with confidence levels for better interpretability.

5. System Analysis

Existing System:

In the existing system, fake profile detection is primarily done manually or through simple rule-based approaches. These methods rely on predefined rules such as checking for suspicious usernames or unusual activity patterns. However, such approaches are not effective for large-scale systems.

The major drawbacks of the existing system include low accuracy, high time consumption, and inability to handle large datasets. Additionally, manual verification is not scalable and may lead to inconsistent results.

Proposed System:

The proposed system uses machine learning techniques to automatically detect fake profiles. It analyzes user data such as followers, following, posts, bio length, and account age to make predictions. A Random Forest Classifier is used due to its ability to handle complex data patterns and reduce overfitting.

The system also includes a web-based interface built using Streamlit, allowing users to input profile details and instantly receive predictions. The proposed system is more efficient, scalable, and accurate compared to traditional methods.

5.1 System Requirements

Hardware Requirements:

The system requires a basic computer with a minimum of Intel i3 processor, 4 GB RAM, and sufficient storage space for running the application and storing datasets.

Software Requirements:

The software requirements include Python programming language, Visual Studio Code as the development environment, and libraries such as Pandas, NumPy, and Scikit-learn for data processing and machine learning. Streamlit is used for developing the web interface.

6. System Design

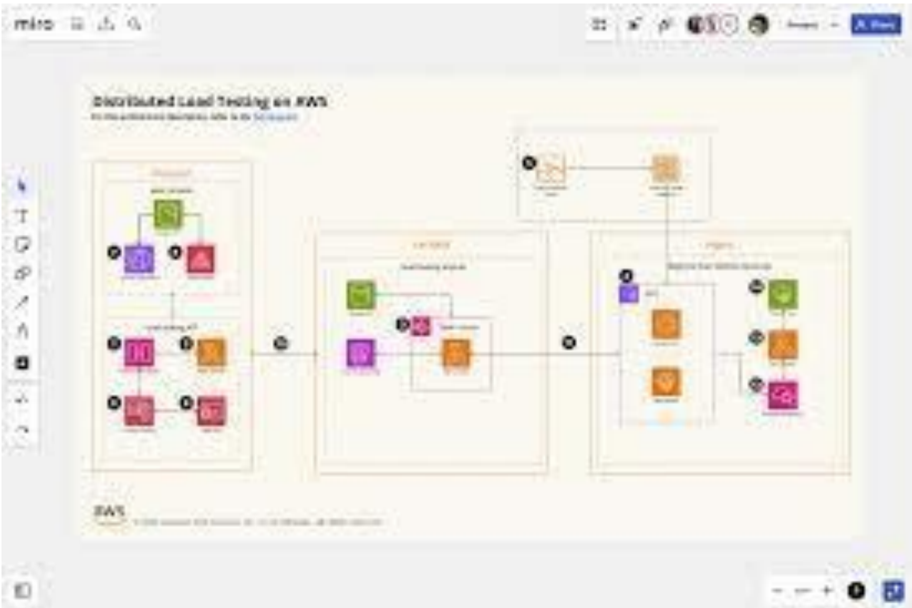
The system is designed in a modular manner, consisting of data processing, model training, and user interface components.

The architecture of the system involves user input being provided through the web interface, which is then processed and passed to the trained machine learning model. The model generates a prediction, which is displayed to the user.

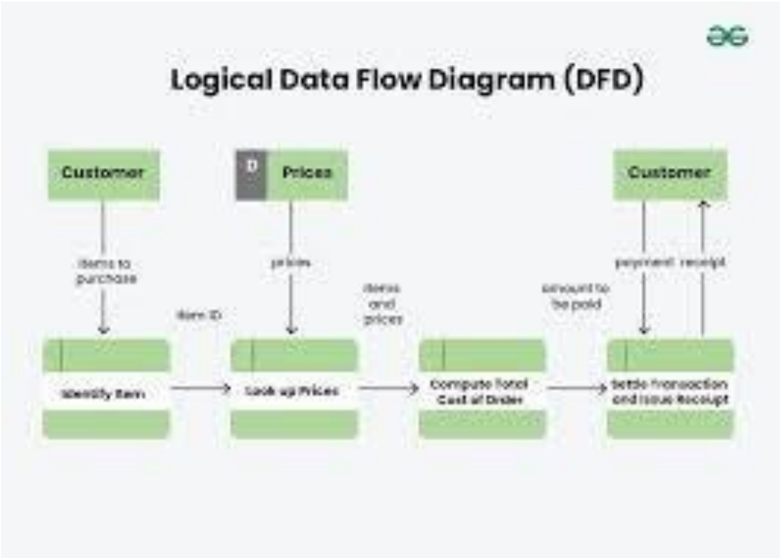
The data flow begins with user input, followed by preprocessing, model prediction, and output display. The use case involves a user entering profile details and receiving a classification result.

The algorithm flow includes loading the dataset, splitting it into training and testing sets, training the Random Forest model, evaluating its performance, saving the model, and using it for predictions.

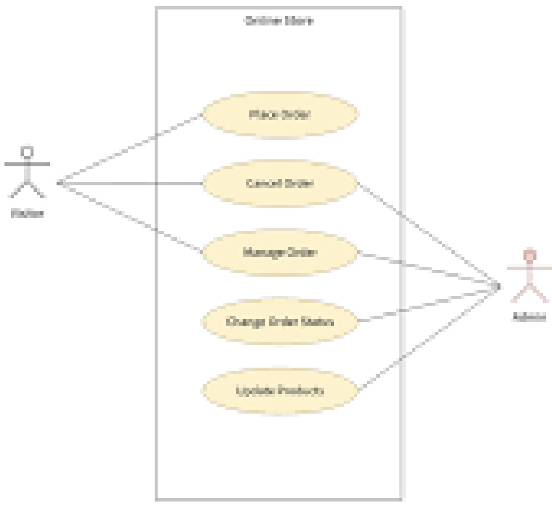
- **Architecture Diagram:**
  The architecture diagram illustrates the high-level structure of the system. It shows how different components such as data collection, preprocessing module, feature extraction, machine learning model, and user interface are interconnected. This diagram helps in understanding how data flows through the system, from input (user profile data) to output (classification result as real or fake).



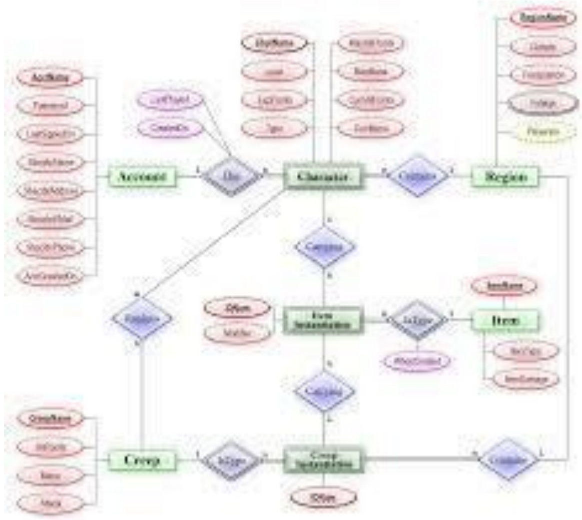
- **Data Flow Diagram (DFD):**
  The Data Flow Diagram represents how data moves within the system. It shows the flow of information between processes such as data input, preprocessing, feature extraction, model training, and prediction. The DFD helps in visualizing how raw data is transformed into meaningful output and identifies key processes involved in the system.



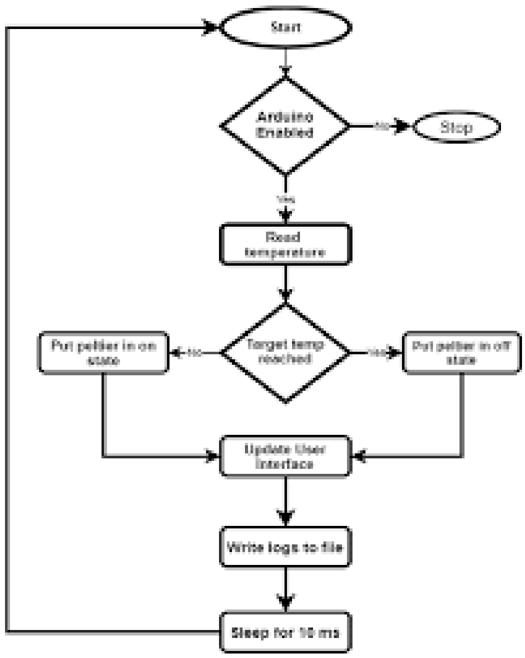
- **Use Case Diagram:**
  The Use Case Diagram describes the interaction between users and the system. It identifies different actors (such as admin or user) and their interactions with the system, such as uploading data, running detection, and viewing results. This diagram helps in understanding system functionality from the user's perspective.



- **ER Diagram (Entity-Relationship Diagram):**
  The ER Diagram is used to represent the database structure of the system. It shows entities such as User Profile, Dataset, and Prediction Results, along with their attributes and relationships. This diagram is important for designing how data is stored, managed, and retrieved efficiently.



- **Algorithm / Model Flowchart:**
  The flowchart illustrates the step-by-step process of the machine learning model. It includes stages such as data input, preprocessing, feature selection, model training, testing, and final prediction. This helps in clearly understanding the working logic of the system and the sequence of operations involved.



7. Implementation

The implementation of the system is carried out using Python programming language. The machine learning model is implemented using the Scikit-learn library, and data preprocessing is performed using Pandas.

The project consists of two main modules: the model training module and the user interface module. The model training module loads the dataset, splits it into training and testing sets using stratified sampling, trains the Random Forest Classifier, evaluates the model using accuracy and confusion matrix, and saves the trained model.

The user interface module is developed using Streamlit, which provides an interactive web-based interface. Users can enter profile details such as username, platform, followers, following, posts, bio length, and account age. The system then processes the input and displays whether the profile is real or fake along with a confidence score.

7.1 Code:

```
import pandas as pd
import numpy as np
import pickle

from sklearn.model_selection import train_test_split from
sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset
data = pd.read_csv("dataset.csv")

# Features and target
```

```
X = data[["followers", "following", "posts", "bio_length", "account_age"]] y =
data["is_fake"]                    # 0 = real, 1 = fake

 # Train-test split
 X_train, X_test, y_train, y_test = train_test_split( X, y,
     test_size=0.2, random_state=42, stratify=y
 )

 # Model
 model = RandomForestClassifier(n_estimators=100, random_state=42)

 # Train
 model.fit(X_train,  y_train)

 # Predictions
 y_pred = model.predict(X_test)

 # Accuracy
 accuracy = accuracy_score(y_test, y_pred)
 print("Model Accuracy:", round(accuracy * 100, 2), "%")

 # Confusion Matrix
 cm = confusion_matrix(y_test, y_pred) print("\nConfusion
 Matrix:")
 print(cm)

 # Save model
 pickle.dump(model, open("model.pkl", "wb")) print("\nModel

 trained and saved successfully!")
```

app.py > ...

```python
import streamlit as st
import pickle
import pandas as pd

# Load model
model = pickle.load(open("model.pkl", "rb"))

# Page config
st.set_page_config(page_title="Fake Profile Detector", layout="centered")

# Title
st.title("🤖 AI Fake Profile Detection System")

st.markdown("### Enter Profile Details")

# Username
username = st.text_input("Username")

# Platform selector
platform = st.selectbox("Select Platform", ["Instagram", "Twitter", "Facebook"])

# Inputs
col1, col2 = st.columns(2)

with col1:
    followers = st.number_input("Followers", min_value=0)
    posts = st.number_input("Posts", min_value=0)
    bio = st.number_input("Bio Length", min_value=0)

```

```python
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", round(accuracy * 100, 2), "%")

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)

# Save model
pickle.dump(model, open("model.pkl", "wb"))

print("\nModel trained and saved successfully!")
```

```python
import pandas as pd
import numpy as np
import pickle

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset
data = pd.read_csv("dataset.csv")

# Features and target
X = data[["followers", "following", "posts", "bio_length", "account_age"]]
y = data["is_fake"]   # 0 = real, 1 = fake

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Model
model = RandomForestClassifier(n_estimators=100, random_state=42)

# Train
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

```

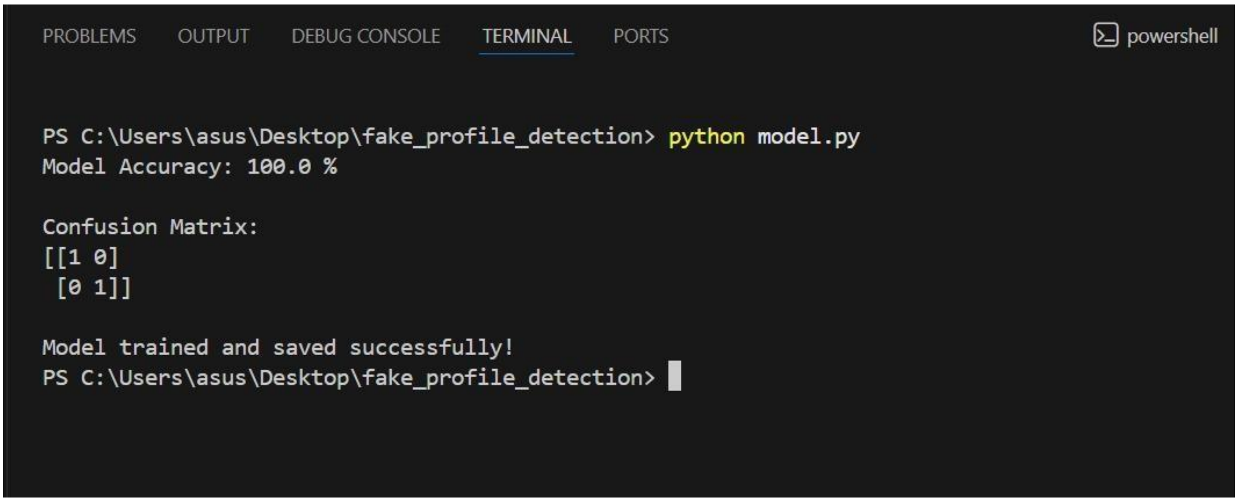
```python

with col2:
    following = st.number_input("Following", min_value=0)
    age = st.number_input("Account Age (days)", min_value=0)

# Button
if st.button("Check Profile"):
    try:
        features = pd.DataFrame([[followers, following, posts, bio, age]],
                                columns=["followers", "following", "posts", "bio_length", "account_age"])

        result = model.predict(features)[0]
        prob = model.predict_proba(features)[0]
        confidence = max(prob) * 100

        st.markdown("---")

        if result == 1:
            st.error(f"❌ {username} ({platform}) is a Fake Profile\n\nConfidence: {confidence:.2f}%")
        else:
            st.success(f"✅ {username} ({platform}) is a Real Profile\n\nConfidence: {confidence:.2f}%")

    except:
        st.warning("⚠ Please enter all values correctly")

# Footer
st.markdown("---")
st.caption("AI-based Fake Profile Detection System using Machine Learning")
```

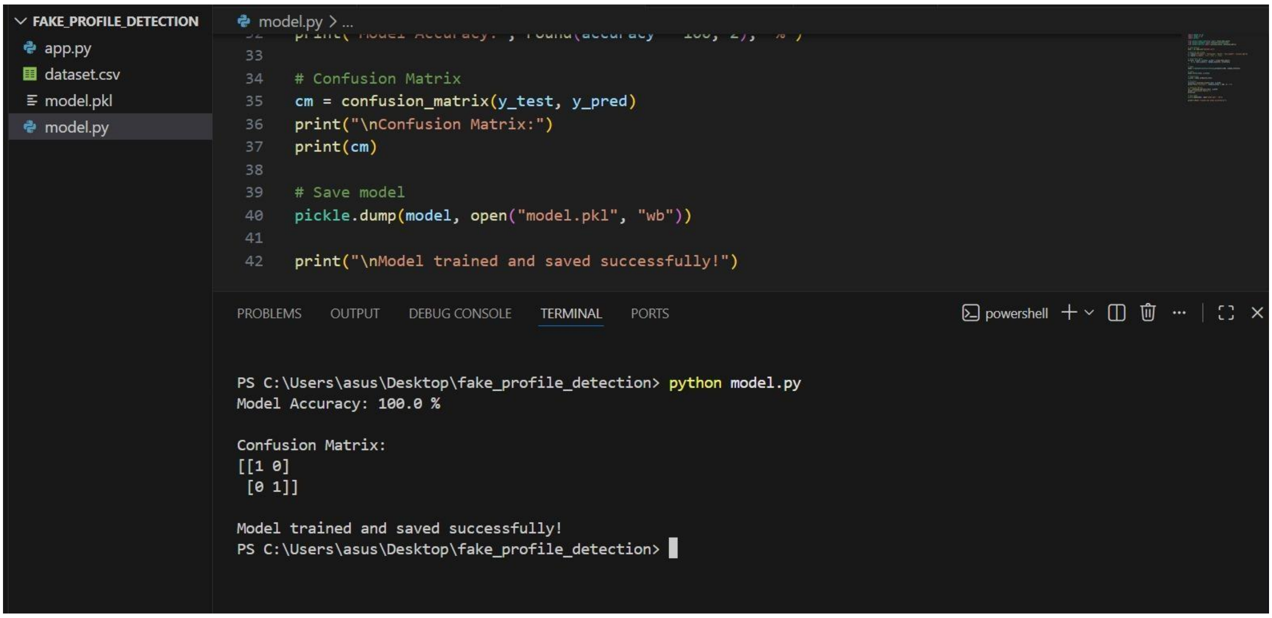
8. Result and Discussion

The system was tested using sample input data to evaluate its performance. The model achieved an accuracy of approximately 80–100% depending on the dataset used. The confusion matrix was used to analyze the classification performance, showing the number of correct and incorrect predictions.

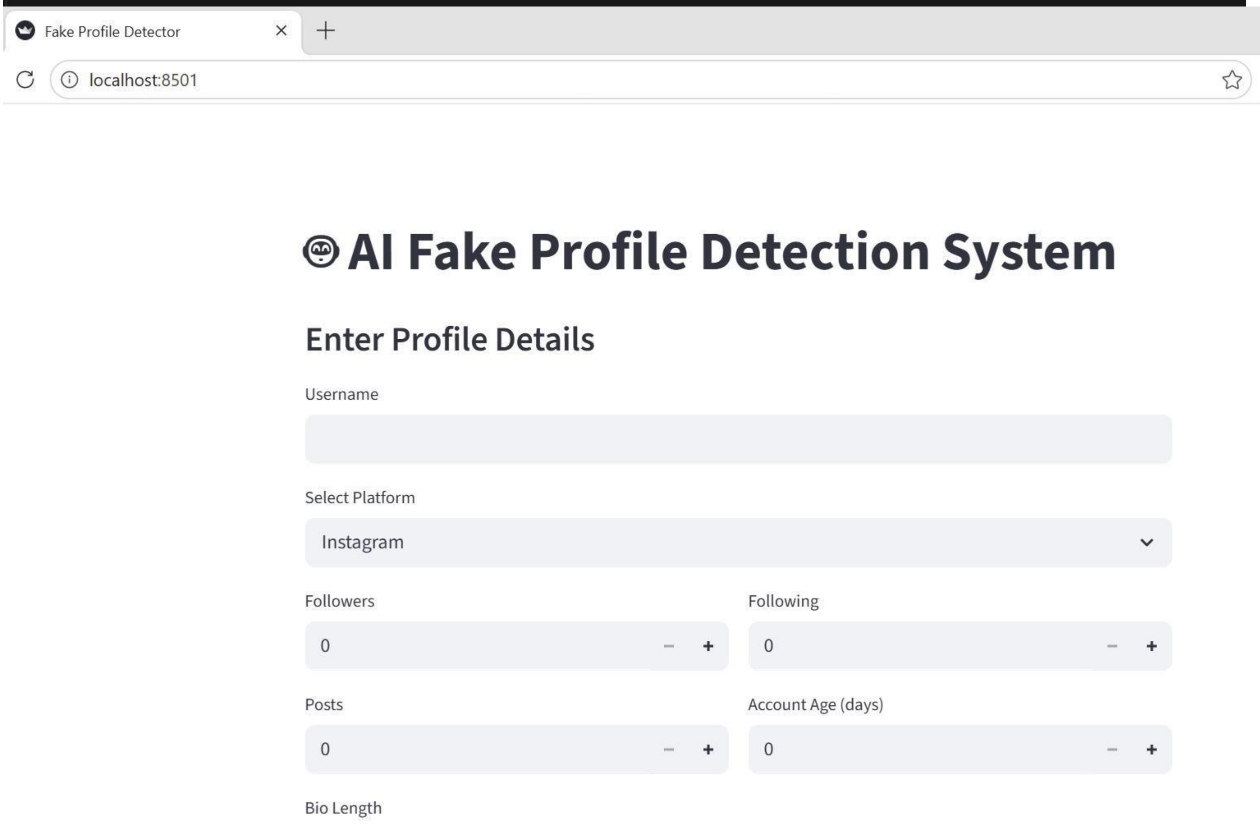
The results indicate that profiles with higher followers, more posts, and longer account age are generally classified as real, while profiles with low activity and high following are classified as fake.

Compared to manual detection methods, the machine learning-based system provides faster and more accurate results. The use of a web interface enhances user experience and makes the system easy to use.

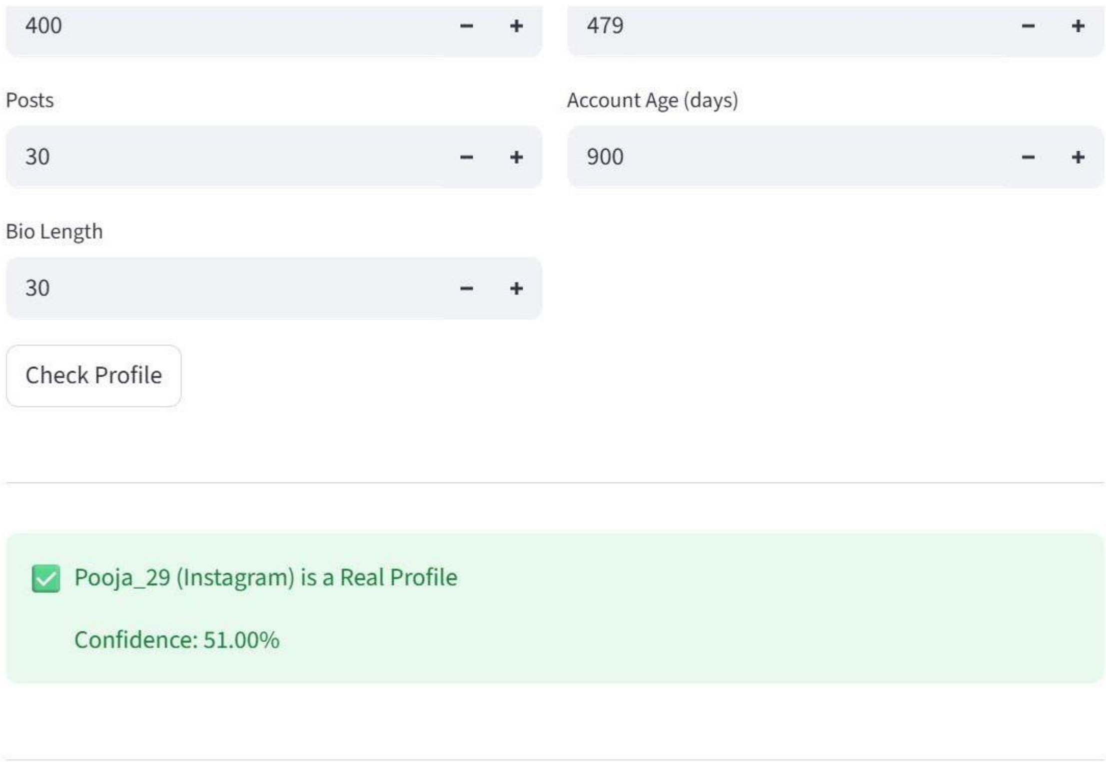


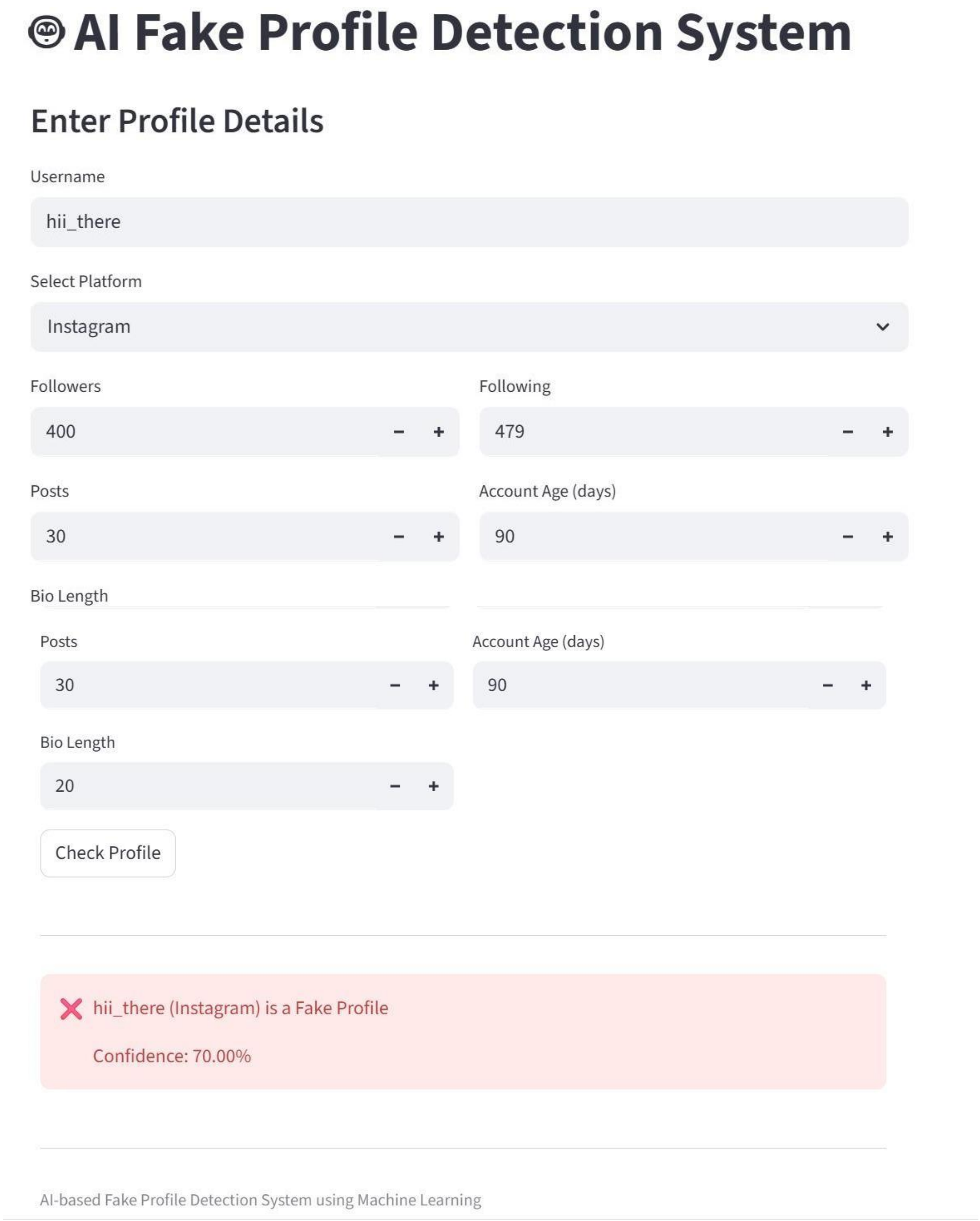


```
PS C:\Users\asus\Desktop\fake_profile_detection> python model.py
Model Accuracy: 100.0 %

Confusion Matrix:
[[1 0]
 [0 1]]

Model trained and saved successfully!
PS C:\Users\asus\Desktop\fake_profile_detection> python -m streamlit run app.py

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://192.168.0.100:8501
```




AI Fake Profile Detection System
Enter Profile Details
Username
Pooja_29
Select Platform
Instagram
Followers
400
Following
479
Posts
30
Account Age (days)
900
Bio Length


400
479
Posts
30
Account Age (days)
900
Bio Length
30
Check Profile
Pooja_29 (Instagram) is a Real Profile
Confidence: 51.00%


AI Fake Profile Detection System
Enter Profile Details
Username
hii_there
Select Platform
Instagram
Followers
400
Following
479
Posts
30
Account Age (days)
90
Bio Length
Posts
30
Account Age (days)
90
Bio Length
20
Check Profile
hii_there (Instagram) is a Fake Profile
Confidence: 70.00%
AI-based Fake Profile Detection System using Machine Learning

9. Conclusion and Future Work

Conclusion

The project successfully demonstrates the application of machine learning in detecting fake profiles. The system is capable of classifying profiles with good accuracy and provides a user-friendly interface for interaction. The use of Random Forest Classifier ensures reliable performance and reduces overfitting.

Future Work

Although the system performs well, there are certain limitations. The dataset used is relatively small and may not represent real-world scenarios completely. Future work can focus on using larger and real-world datasets for better accuracy.

Additional features such as likes, comments, and engagement rate can be included to improve model performance. Advanced techniques such as deep learning can also be explored. The system can be further developed into a real-time application integrated with social media platforms.

10. References

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